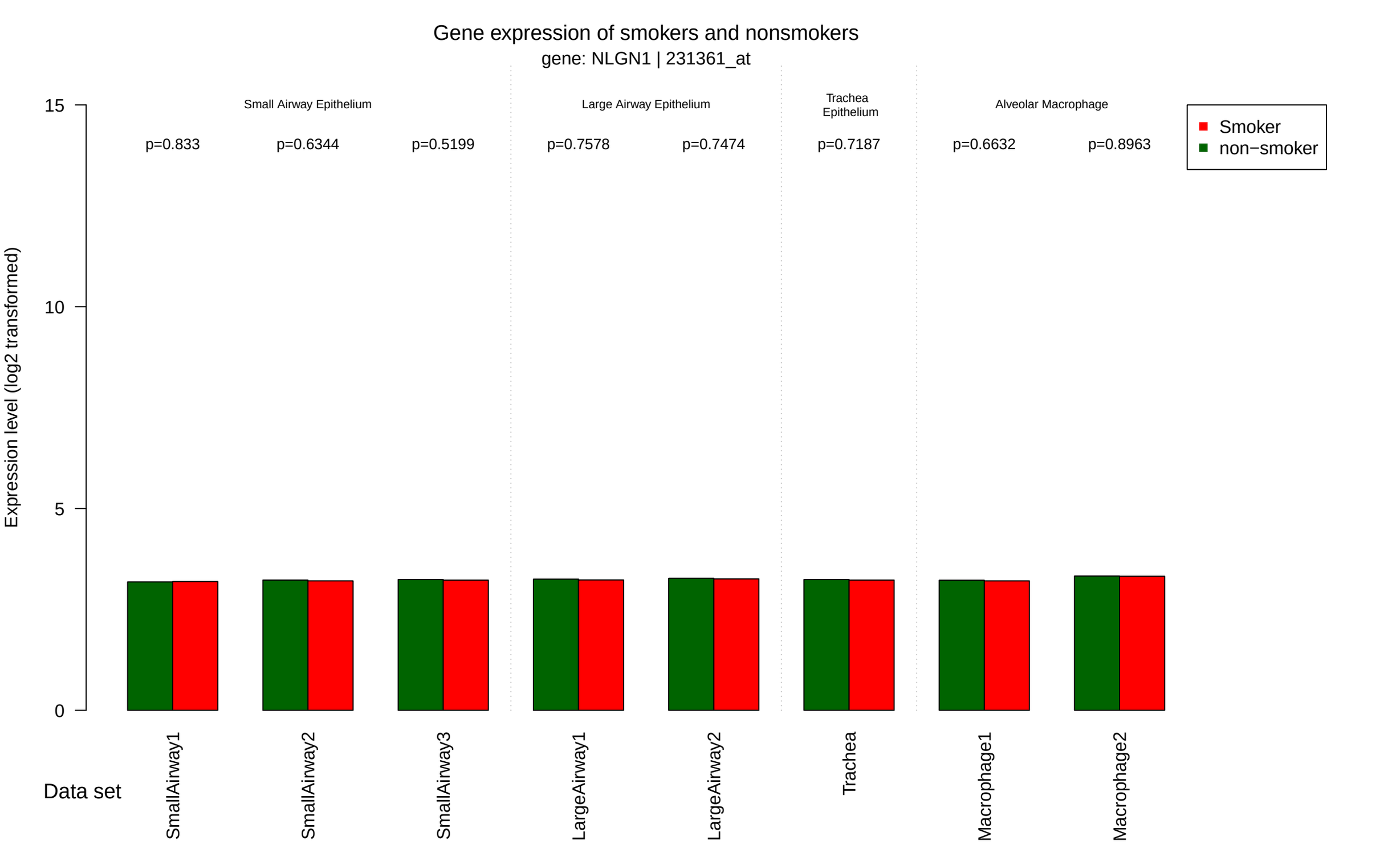
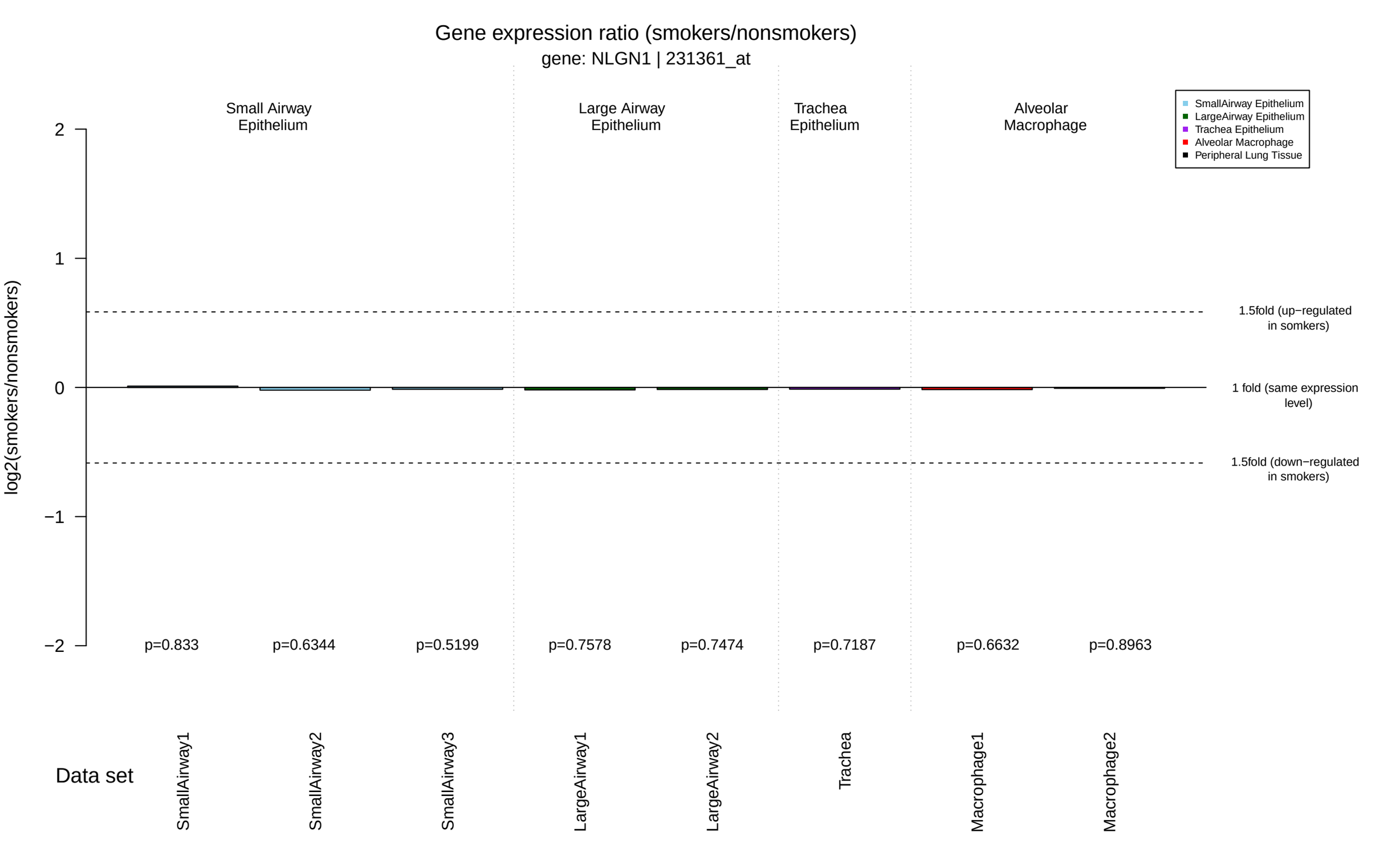
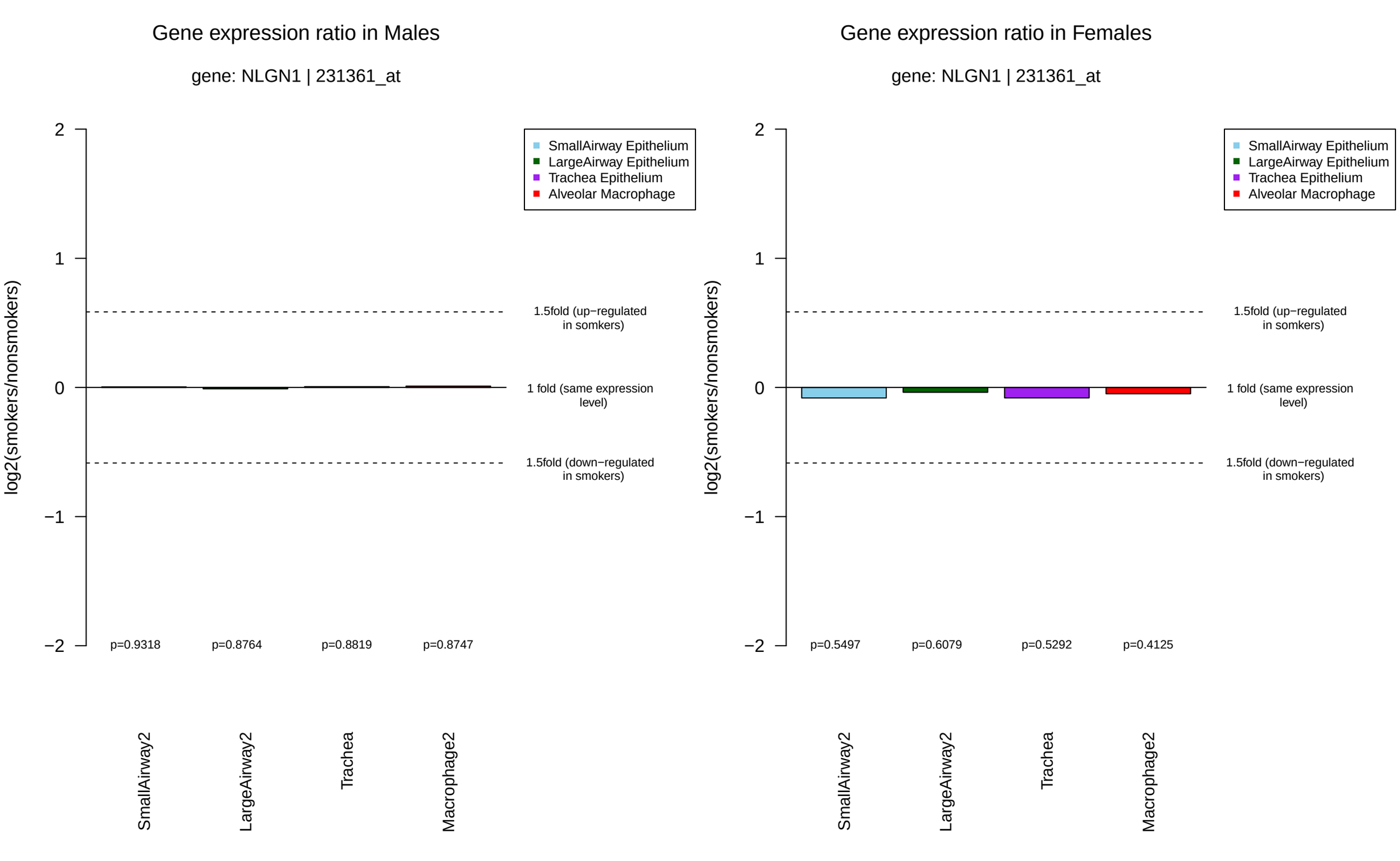
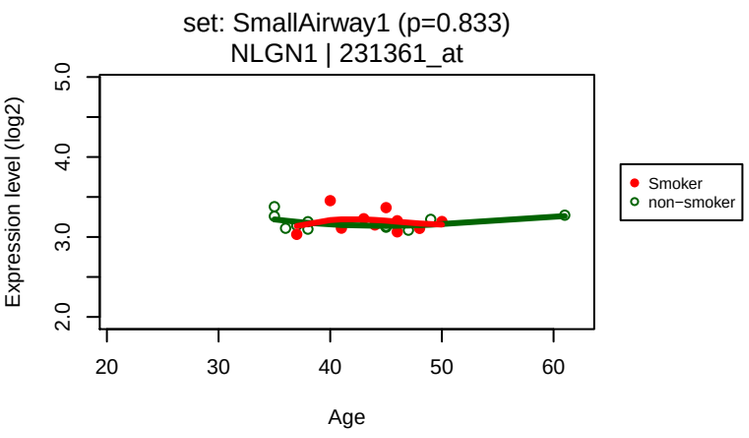
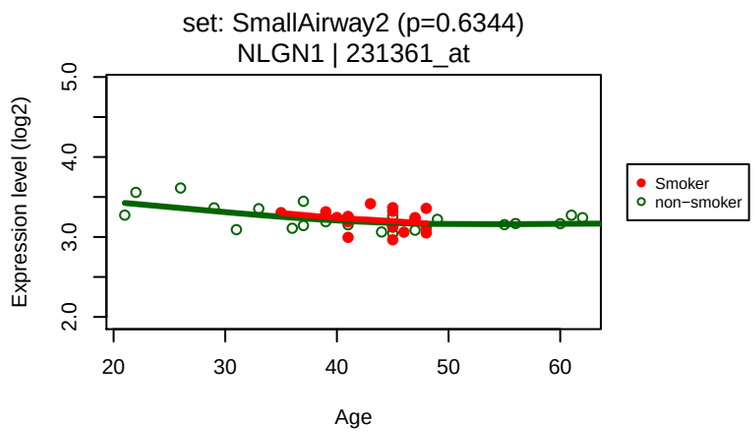
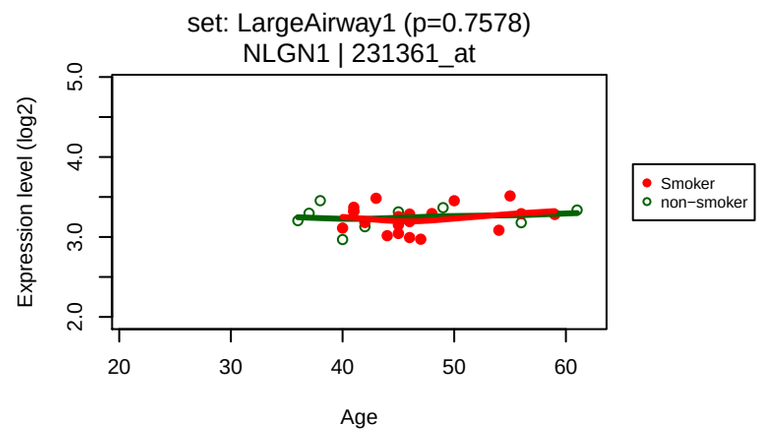

Gene expression of smokers and nonsmokers
gene: NLGN1 | 231361_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.833
p=0.6344
p=0.5199
p=0.7578
p=0.7474
p=0.7187
p=0.6632
p=0.8963
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

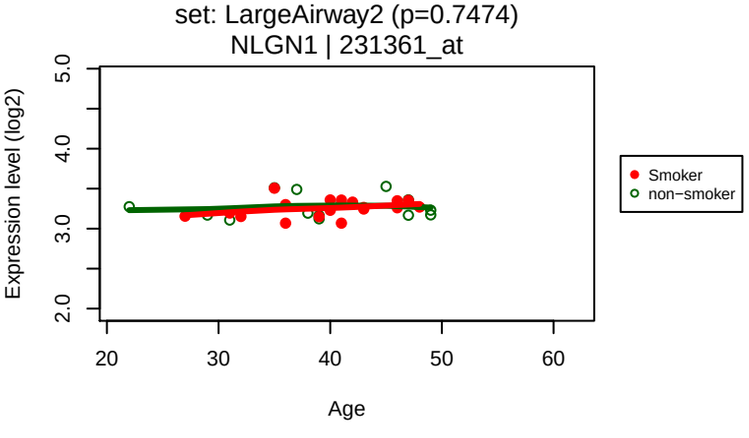

Gene expression ratio (smokers/nonsmokers)
gene: NLGN1 | 231361_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.833
p=0.6344
p=0.5199
p=0.7578
p=0.7474
p=0.7187
p=0.6632
p=0.8963
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

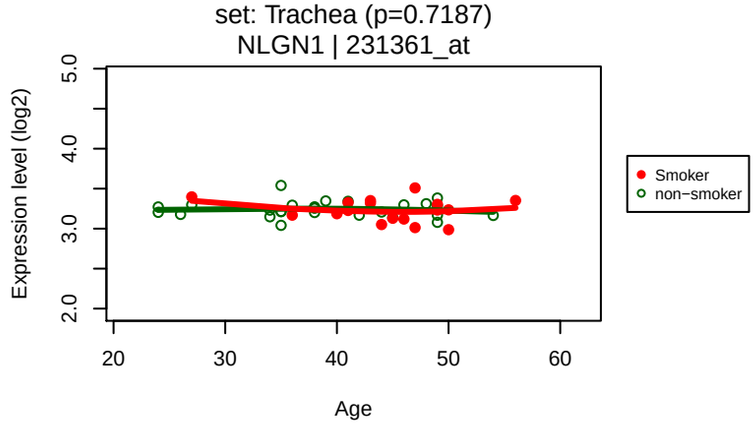

Gene expression ratio in Males
gene: NLGN1 | 231361_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9318
p=0.8764
p=0.8819
p=0.8747
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NLGN1 | 231361_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5497
p=0.6079
p=0.5292
p=0.4125
SmallAirway2
LargeAirway2
Trachea
Macrophage2

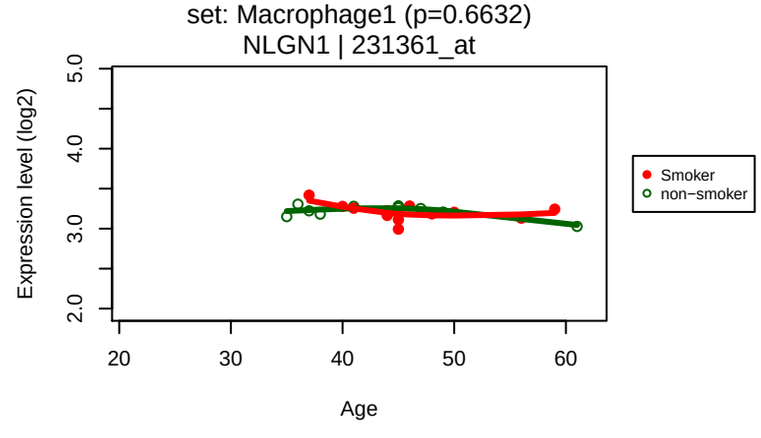

set: SmallAirway1 (p=0.833)
NLGN1 | 231361_at
Expression level (log2)
Age
Smoker
non−smoker

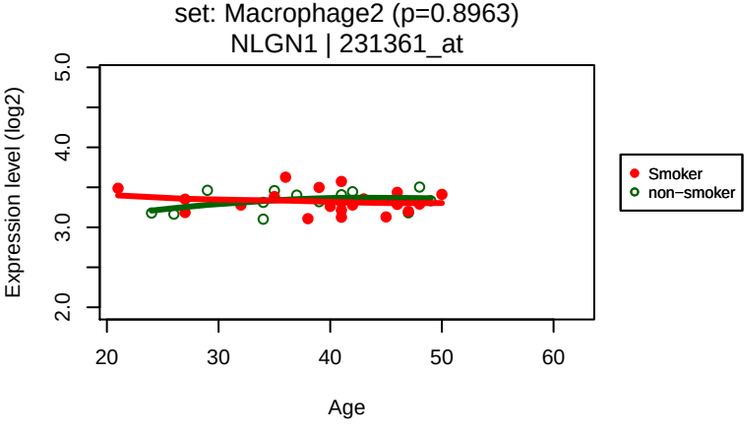

set: SmallAirway2 (p=0.6344)
NLGN1 | 231361_at
Expression level (log2)
Age
Smoker
non−smoker

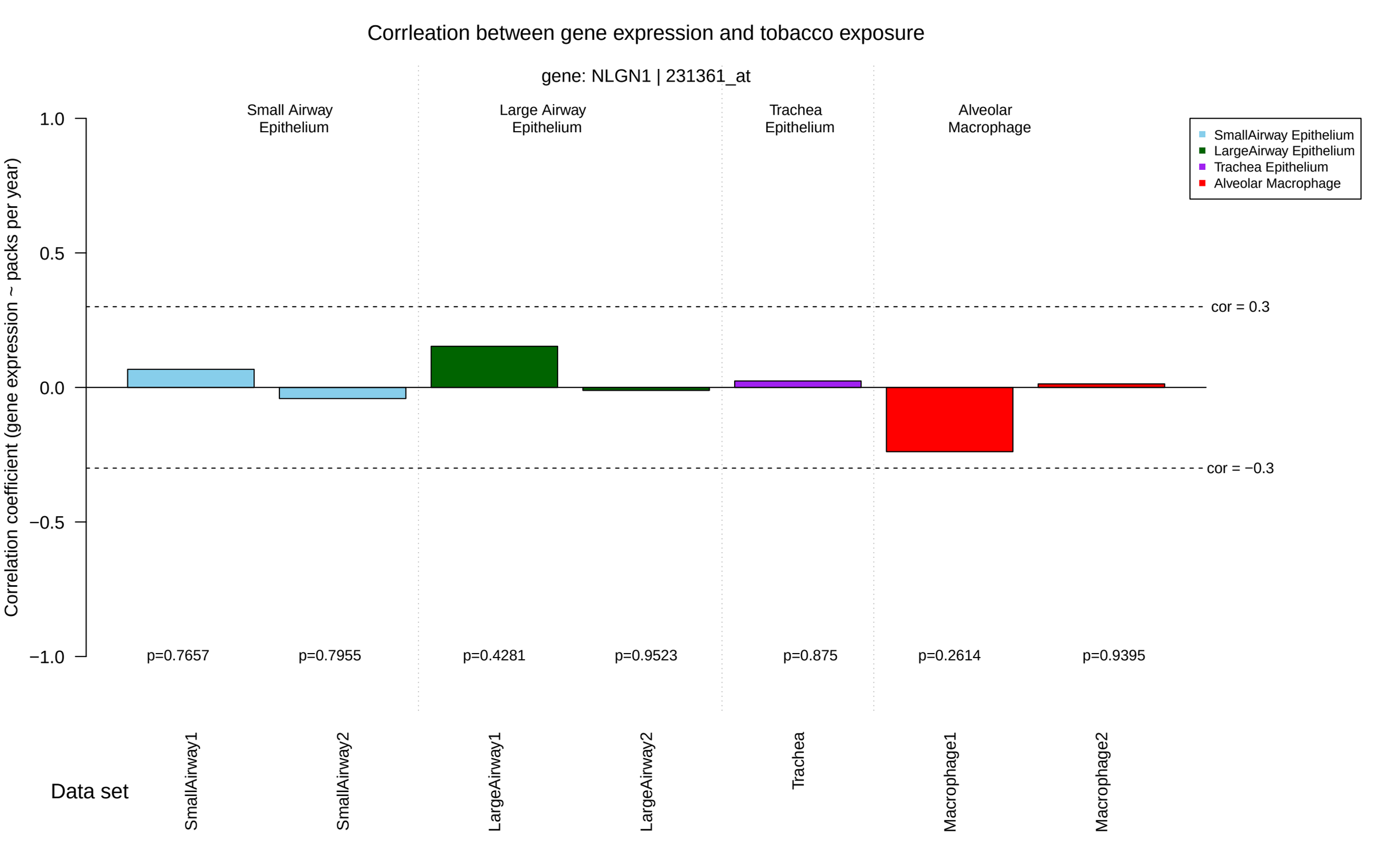

set: LargeAirway1 (p=0.7578)
NLGN1 | 231361_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7474)
NLGN1 | 231361_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7187)
NLGN1 | 231361_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6632)
NLGN1 | 231361_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8963)
NLGN1 | 231361_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NLGN1 | 231361_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7657
p=0.7955
p=0.4281
p=0.9523
p=0.875
p=0.2614
p=0.9395
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2